Foundations of the Calculus Transform

The 8S Penteract–S^3 edition carried to the continuum — definitions, master theorems Θ-A…Θ-J, differential lemmas D1…D14, and proofs

Contents

Foundations of the Calculus Transform **1**

0. The tower 1
1. Standing analytic objects and hypotheses 2
2. The calculus transform operator 2
3. Master theorems 3
4. The fourteen differential lemmas 8
5. The per-entry proof schema 11
6. Scope and honest limits 11

Foundations of the Calculus Transform

This document is the proof apparatus for the calculus edition of the LKVexa Compendium. The 8S edition replaced each linear-algebraic statement by an algebraic object on a finite lattice: a graph Laplacian, a projector, a kernel matrix, a gap function. The calculus edition takes those objects to the continuum, where they become differential and integral operators, energy functionals, flows, and variational principles.

The transform is not a change of vocabulary. Each 8S object has a *limit*, and the limit is a theorem with hypotheses and an error rate. Theorem Θ-G is the central one: the 8S Laplacian $\mathcal{L}_{8S}$ converges, as the kernel bandwidth shrinks, to a second-order elliptic operator in divergence form, with relative error $O(\sigma^2)$. Everything downstream of a spectrum inherits that convergence.

Every per-entry proof in `md-cal/01`–`md-cal/80` is a six-step instantiation of material established here.

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0. The tower

Three editions now stand in a chain, each containing the previous:

$$\underbrace{\mathcal{L}}_{\textbf{LA edition}} \xrightarrow{\mathsf{T}_{8S}} \underbrace{\mathsf{T}_{8S}[\mathcal{L}]}_{\textbf{8S edition}} \xrightarrow{\mathsf{T}_{\mathrm{cal}}} \underbrace{\mathsf{T}_{\mathrm{cal}}[\mathsf{T}_{8S}[\mathcal{L}]]}_{\textbf{calculus edition}}$$

with retractions running the other way:

$$\mathsf{T}_{8S}^{(0)} = \mathrm{id} \quad \text{(8S Theorem J)}, \qquad S_\ell \circ \mathsf{T}_{\mathrm{cal}} = \mathsf{T}_{8S} + O(\ell^2) \quad \text{(Θ-J below)},$$

where S_ℓ is restriction to the lattice. Composing, every calculus image returns its own input equation:

$$\mathsf{T}_{8S}^{(0)} \circ S_\ell \circ \mathsf{T}_{\mathrm{cal}}[\mathsf{T}_{8S}[\mathcal{L}]] \;=\; \mathcal{L}.$$

That two-step retraction is step (Q5) of every per-entry proof, and it is what makes the calculus edition an extension of the corpus rather than a replacement of it.

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1. Standing analytic objects and hypotheses

Base domain. $\mathcal{C} = [-1,1]^5$ with coordinates c, carrying the constant metric $G_5 \succ 0$ and the norm $\|v\|_{G_5}^2 = v^\top G_5 v$. The 8S lattice $\mathcal{C}_M = \{\ell n : n \in \mathbb{Z}^5, \|n\|_1 \le M\}$ is replaced by $\mathcal{C}$ together with a **sampling density** $p \in C^2(\mathcal{C})$, $p > 0$, normalised to $\int_{\mathcal{C}} p\, dc = 1$. The lattice is recovered by S_ℓ, restriction to $\mathcal{C}_M$.

Kernel and bandwidth. $W_\sigma(c,c') = \exp(-\|c-c'\|_{G_5}^2/2\sigma_c^2)\exp(-d_{\mathrm{J}}(c,c')^2/2\sigma_{\mathrm{J}}^2)$, with $\sigma := \sigma_c$ the bandwidth taken to zero. Where the tag factor is present it is assumed C^1 in its arguments and bounded away from zero, so it contributes a smooth positive multiplier absorbed into p; this is stated wherever used.

Degree and averaging operator.

$$D_\sigma(c) = \int_{\mathcal{C}} W_\sigma(c,c')p(c')\,dc', \qquad (\mathcal{A}_\sigma f)(c) = \frac{1}{D_\sigma(c)}\int_{\mathcal{C}} W_\sigma(c,c')f(c')p(c')\,dc'.$$

Continuum 8S operator.

$$\mathcal{L}_\sigma := \frac{1}{\sigma^2}(I - \mathcal{A}_\sigma), \qquad \boxed{\mathcal{L}_\star f := -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla f)}$$

Gate. $p_\star = \sigma(h)$ becomes a field $\gamma(c) \in (0,1)$; Γ becomes multiplication by γ. Magnetisation $m = 2\gamma - 1 \in (-1,1)$.

Radius and fibre. $r \in C^1(\mathcal{C})$, $r > 0$ on the admissible region; the total space is the sphere bundle

$$X_8 = \{(c,y) \in \mathcal{C}\times\mathbb{R}^4 : \|y\|_2 = r(c)\}, \qquad \Phi(c,u) = (c,\ r(c)u),\ u \in S^3,$$

with the metric $\mathcal{G}_8 = \mathrm{blkdiag}(G_5, I_4)$ on the ambient $\mathbb{R}^9$.

Projection. $\Pi(\tau) = [\Pi_5(\tau)\ \Psi_4(\tau)] \in \mathbb{R}^{3\times 9}$, C^1 in τ, with the standing normalisation (N) of the 8S edition: $\Pi\Pi^\top = I_3$, $\Pi_5\Pi_5^\top = I_3$, $G_5 \succeq I_5$, $\hat{r} \ge \lambda$.

Regularity hypotheses. Three are used, and each per-entry proof names the ones it needs:

- **(R1)** $f \in C^4(\mathcal{C})$ and $p \in C^2(\mathcal{C})$, $p > 0$ — needed for the $O(\sigma^2)$ rate in Θ-G. Under $f \in C^3$ only, the rate degrades to $O(\sigma)$.
- **(R2)** the evaluation point lies in the interior at distance $\gg \sigma$ from $\partial\mathcal{C}$ — needed because the kernel is truncated at the boundary; see Remark Θ-G .
- **(R3)** $r \in C^1$ and $r > 0$ — needed for the bundle to be a smooth manifold and for Θ-E.

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2. The calculus transform operator

Definition 2.1 (continuum limit of a lattice object). Let X_ℓ be an 8S object defined on $\mathcal{C}_M$. Its **calculus image** is the operator, functional, or form $X_\star$ on $\mathcal{C}$ such that $S_\ell X_\star = X_\ell + O(\ell^2)$ in the sense of Riemann sums, and $X_\star = \lim_{\sigma\to 0} X_\sigma$ where the σ-family is the kernel-smoothed version of X_ℓ.

Definition 2.2 (the calculus transform). For an 8S image $\mathsf{T}_{8S}[\mathcal{L}]$ presented as a relation among the objects $\{\mathcal{L}_{8S}, \Gamma, W, \mathcal{R}, \Pi, \eta_{\mathrm{ind}}, \delta^{(8)}, g^{(\bullet)}, \Delta_{8S}\}$, its **calculus image** $\mathsf{T}_{\mathrm{cal}}[\mathsf{T}_{8S}[\mathcal{L}]]$ is obtained by the substitutions

$$
\begin{aligned}
\textstyle\sum_j W_{ij}(\cdot)_j &\longmapsto \int_{\mathcal{C}} W_\sigma(c,c')(\cdot)(c')\,p(c')\,dc' && \text{(sum to integral)}\\
\mathcal{L}_{8S} &\longmapsto \mathcal{L}_\star = -\tfrac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot) && (\Theta\text{-G})\\
W_\sigma &\longmapsto e^{t\mathcal{A}_G},\ \ \mathcal{A}_G = \nabla\cdot(G_5^{-1}\nabla\,\cdot),\ t=\sigma^2/2 && (\Theta\text{-D})\\
\Gamma &\longmapsto \text{multiplication by } \gamma,\ \ \tfrac{d\gamma}{dh}=\gamma(1-\gamma) && (\Theta\text{-B})\\
\text{quadratic form} &\longmapsto \text{Dirichlet energy } \mathcal{E}[\phi]=\tfrac{1}{2}\textstyle\int\|\nabla\phi\|^2_{G_5^{-1}}p^2dc && (\Theta\text{-C})\\
\delta^{(8)} &\longmapsto d_g,\ \text{the geodesic distance of } ds^2 \text{ on } X_8 && (\Theta\text{-E})\\
\text{rank / null space} &\longmapsto \dim\ker,\ \text{ind},\ H^0 && (\Theta\text{-A},\Theta\text{-G})\\
\max\Delta_{8S} &\longmapsto \delta\Delta_{8S}=0 \text{ with KKT multipliers} && (\Theta\text{-H})\\
\eta_{\text{ind}} &\longmapsto \text{Gram-determinant ratio; } L^2 \text{ projection residual} && (\Theta\text{-A}).
\end{aligned}
$$

Definition 2.3 (differential signature). The **order** $\mathrm{ord}(\mathcal{L})$ of an input equation is 0 if it contains no derivative or integral, 1 if it contains a first derivative or a single integral, and 2 if it contains a second derivative, a Laplacian, or a double integral. The signature is recorded in step (Q1) of each per-entry proof and governs which regularity hypothesis is invoked: an equation already of order ≥ 1 needs no smoothing to have a calculus image, and for it the transform acts near-identically on the differential part.

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3. Master theorems

Theorem Θ-A — the independence gate as an L^2 projection and a Gram ratio

Let $\mathcal{V}=\mathrm{span}\{v_1,\dots,v_4\}\subset L^2(\mathcal{C},p\,dc)$ be closed with $C_4=[v_1\ \cdots\ v_4]$ of full column rank, and let $e\in L^2$.

1. There is a unique $P_{\mathcal{V}}e\in\mathcal{V}$ minimising $\|e-v\|_{L^2}$ over $v\in\mathcal{V}$, characterised by the variational condition $\langle e-P_{\mathcal{V}}e,\ v\rangle=0$ for all $v\in\mathcal{V}$.
2. $\|w\|^2=\dfrac{\det\mathrm{Gr}(C_4,e)}{\det\mathrm{Gr}(C_4)}$, where $w=e-P_{\mathcal{V}}e$ and $\mathrm{Gr}(X)_{ab}=\langle x_a,x_b\rangle$.
3. $\eta_{\text{ind}}=\dfrac{\det\mathrm{Gr}(C_4,e)}{\det\mathrm{Gr}(C_4)\,(\|e\|^2+\varepsilon)}$, and $\nabla_e\|w\|^2=2w$.
4. $\eta_{\text{ind}}>0$ iff $\{v_1,\dots,v_4,e\}$ is linearly independent, iff the Gram determinant is nonzero.

Proof. (1) is the Hilbert projection theorem: $\mathcal{V}$ is a closed convex subset of a Hilbert space, so the nearest point exists and is unique; the first-order condition $\frac{d}{ds}\|e-Pe-sv\|^2\big|_{s=0}=-2\langle e-Pe,v\rangle=0$ gives the orthogonality, and conversely orthogonality implies minimality by Pythagoras.

(2) Write the block Gram matrix

$$\mathrm{Gr}(C_4,e)=\begin{pmatrix}\mathrm{Gr}(C_4) & C_4^*e\\ e^*C_4 & \langle e,e\rangle\end{pmatrix}.$$

Since $\mathrm{Gr}(C_4)$ is invertible (full column rank), the Schur determinant formula gives

$$\det\mathrm{Gr}(C_4,e)=\det\mathrm{Gr}(C_4)\cdot\Big(\langle e,e\rangle-e^*C_4\,\mathrm{Gr}(C_4)^{-1}C_4^*e\Big)=\det\mathrm{Gr}(C_4)\cdot\langle e,(I-P_{\mathcal{V}})e\rangle,$$

because $C_4\mathrm{Gr}(C_4)^{-1}C_4^*$ is exactly $P_{\mathcal{V}}$. Finally $\langle e,(I-P)e\rangle=\langle(I-P)e,(I-P)e\rangle=\|w\|^2$ using that $I-P$ is a self-adjoint idempotent.

(3) is (2) divided by $\|e\|^2+\varepsilon$; the gradient is $\nabla_e\langle e,(I-P)e\rangle=2(I-P)e=2w$ by symmetry of $I-P$.

(4) A Gram determinant vanishes iff its vectors are dependent, which is the classical criterion; combine with (2). ■

Remark Θ-A . Part (3) upgrades the 8S gate from a test to a *variational* object: w is the gradient of the squared residual, hence the steepest-ascent direction of novelty. Part (2) is the calculus of volumes: $\det\mathrm{Gr}$ is the squared 4-volume of the parallelepiped spanned by C_4, so η_{ind} is a normalised ratio of a 5-volume to a 4-volume — a height.

Theorem Θ-B — the gate is the flow of the logistic vector field

Let $\gamma = \sigma(h) = (1 + e^{-h})^{-1}$.

1. γ satisfies the logistic ODE $\dfrac{d\gamma}{dh} = \gamma(1-\gamma)$ with $\gamma(0) = \frac{1}{2}$, and is the time-h flow of the vector field $V(\gamma) = \gamma(1-\gamma)$ on $(0,1)$.
2. $\displaystyle\int \frac{d\gamma}{\gamma(1-\gamma)} = \log\frac{\gamma}{1-\gamma} = h$, so the logit is the inverse flow time.
3. $\gamma' = \gamma(1-\gamma) \le \frac{1}{4}$ with equality only at $\gamma = \frac{1}{2}$; $\gamma'' = \gamma(1-\gamma)(1-2\gamma)$, so $h = 0$ is the unique inflection.
4. The Fisher gate of the 8S edition is $\Gamma^2_\star = \gamma'$: the derivative of the gate *is* its Fisher information.

Proof. (1) Differentiate: $\sigma'(h) = e^{-h}(1+e^{-h})^{-2} = \sigma(h)(1-\sigma(h))$. The vector field $V(\gamma) = \gamma(1-\gamma)$ is locally Lipschitz on $(0,1)$, so Picard–Lindelöf gives a unique maximal solution through $\gamma(0) = 1/2$; σ is such a solution and is defined for all h, hence is the flow. (2) Separate variables and use partial fractions $\frac{1}{\gamma(1-\gamma)} = \frac{1}{\gamma} + \frac{1}{1-\gamma}$; integrate to $\log\gamma - \log(1-\gamma)$. (3) Maximise $\gamma(1-\gamma)$ on $(0,1)$; differentiate again for γ''. (4) For a Bernoulli variable with mean γ and natural parameter h, the Fisher information is $\gamma(1-\gamma)$, which by (1) equals $d\gamma/dh$. ■

Theorem Θ-C — congruence becomes the Dirichlet energy; inertia becomes a sign

Define the Dirichlet energy and its gated form

$$\mathcal{E}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \mathcal{E}_\gamma[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla(\gamma\phi)\|^2_{G_5^{-1}}\, p^2\, dc.$$

1. $\mathcal{E}$ is the quadratic form of $\mathcal{L}_\star$: for $\phi, \psi \in C^2$ with ϕ or ψ vanishing on $\partial\mathcal{C}$, or with the Neumann condition $\partial_\nu\phi = 0$,
$$\langle \mathcal{L}_\star\phi, \psi\rangle_{p^2} = \tfrac{1}{2}\int \nabla\phi^\top G_5^{-1}\nabla\psi\; p^2\, dc, \qquad \langle \mathcal{L}_\star\phi, \phi\rangle_{p^2} = \mathcal{E}[\phi] \;\ge 0.$$
2. $\mathcal{L}_\star$ is self-adjoint and positive semidefinite on $L^2(p^2dc)$, with $\ker\mathcal{L}_\star$ = locally constant functions.
3. The first variation of $\mathcal{E}$ gives the Euler–Lagrange equation $\mathcal{L}_\star\phi = 0$; with the constraint $\|\phi\|_{p^2} = 1$ it gives $\mathcal{L}_\star\phi = \mu\phi$.

Proof. (1) Write $\langle\mathcal{L}_\star\phi,\psi\rangle_{p^2} = -\frac{1}{2}\int\nabla\cdot(p^2G_5^{-1}\nabla\phi)\,\psi\,dc$, the weight p^2 cancelling against $1/p^2$. Integrate by parts (divergence theorem): the boundary term is $-\frac{1}{2}\oint\psi\,p^2\,(G_5^{-1}\nabla\phi)\cdot\nu\,dS$, which vanishes under either stated boundary condition, leaving $\frac{1}{2}\int\nabla\phi^\top G_5^{-1}\nabla\psi\,p^2dc$. Setting $\psi = \phi$ and using $G_5^{-1} \succ 0$ gives nonnegativity.

(2) Self-adjointness is the symmetry of the bilinear form just derived; semidefiniteness is the same computation. If $\mathcal{E}[\phi] = 0$ then $\nabla\phi = 0$ almost everywhere (as $G_5^{-1} \succ 0$, $p > 0$), so ϕ is constant on each connected component.

(3) $\frac{d}{ds}\mathcal{E}[\phi + s\psi]\big|_0 = \int\nabla\phi^\top G_5^{-1}\nabla\psi\,p^2dc = 2\langle\mathcal{L}_\star\phi,\psi\rangle_{p^2}$; vanishing for all admissible ψ forces $\mathcal{L}_\star\phi = 0$ by the fundamental lemma of the calculus of variations. With the normalisation constraint, a Lagrange multiplier μ appears, giving the eigenvalue equation. ■

Remark Θ-C . Sylvester's law of inertia, the 8S edition's Theorem C, has no dimension-counting analogue in infinite dimensions; what survives is the *sign*, and it survives exactly: $\mathcal{E}[\phi] \ge 0$ with equality only on constants. Statements of the form "this form is positive definite" transfer; statements of the form "this matrix has ν negative eigenvalues" transfer only when the operator has discrete spectrum, which on the compact $\mathcal{C}$ with Neumann conditions it does.

Theorem Θ-D — the coupling kernel is the heat semigroup

Let $\mathcal{A}_G = \nabla\cdot(G_5^{-1}\nabla\,\cdot\,)$ on $\mathbb{R}^5$ and let

$$k_t(x) = (4\pi t)^{-5/2}\det(G_5)^{1/2}\exp\Big(-\frac{x^\top G_5 x}{4t}\Big).$$

1. $\partial_t k_t = \mathcal{A}_G k_t$ and $k_t \to \delta_0$ weakly as $t \downarrow 0$; $\int k_t = 1$.
2. $k_s * k_t = k_{s+t}$ (Chapman–Kolmogorov), so $\{k_t\}$ is a convolution semigroup and W_σ is, up to normalisation, k_t with $t = \sigma_c^2/2$.

3. $\widehat{k_t}(\xi) = \exp(-t\,\xi^\top G_5^{-1}\xi)$, so the symbol of $e^{t\mathcal{A}_G}$ is a Gaussian and the symbol of $\mathcal{A}_G$ is $-\xi^\top G_5^{-1}\xi$.
4. $\lim_{\sigma\to 0} \dfrac{2}{\sigma^2}(\mathcal{A}_\sigma - I) = \mathcal{A}_G$ at constant density.

Proof. (1) Substitute $y = G_5^{1/2}x$, under which $\mathcal{A}_G$ becomes the ordinary Laplacian Δ_y and k_t becomes the standard heat kernel $(4\pi t)^{-5/2}e^{-\|y\|^2/4t}$, whose heat-equation property is the classical computation

$$\partial_t(t^{-5/2}e^{-\|y\|^2/4t}) = t^{-5/2}e^{-\|y\|^2/4t}\Big(-\tfrac{5}{2t} + \tfrac{\|y\|^2}{4t^2}\Big) = \Delta_y(t^{-5/2}e^{-\|y\|^2/4t}).$$

The Jacobian factor $\det(G_5)^{1/2}$ makes $\int k_t\,dx = 1$. Weak convergence to δ_0 is the standard approximate-identity argument: $k_t \geq 0$, unit mass, and mass concentrating on $\|x\| \leq \eta$ for any $\eta > 0$.

(2) In the y variables this is the Gaussian convolution identity, verified on the Fourier side by (3): $\widehat{k_s}\widehat{k_t} = e^{-(s+t)\xi^\top G_5^{-1}\xi} = \widehat{k_{s+t}}$, and the Fourier transform is injective on L^1.

(3) The Fourier transform of a Gaussian is a Gaussian: for $A \succ 0$, $\int e^{-x^\top Ax}e^{-i\xi\cdot x}dx = \pi^{5/2}\det(A)^{-1/2}e^{-\xi^\top A^{-1}\xi/4}$. Apply with $A = G_5/4t$.

(4) At constant density $\mathcal{A}_\sigma$ is convolution with the normalised kernel, i.e. $e^{t\mathcal{A}_G}$ with $t = \sigma^2/2$; then $\frac{1}{t}(e^{t\mathcal{A}_G} - I) \to \mathcal{A}_G$ strongly on the domain of the generator, which is the definition of the generator of a C_0-semigroup. ■

Theorem Θ-E — the bundle metric, its volume form, and fibre integration

Assume (R3). Let $\Phi(c,u) = (c, r(c)u)$ embed X_8 in $(\mathbb{R}^9, \mathcal{G}_8)$.

1. The induced Riemannian metric is

$$ds^2 = dc^\top(G_5 + \nabla r\,\nabla r^\top)dc + r^2\,d\Omega_3^2,$$

   where $d\Omega_3^2$ is the round metric on the unit S^3.
2. $\det(G_5 + \nabla r \nabla r^\top) = \det(G_5)(1 + \|\nabla r\|^2_{G_5^{-1}})$.
3. The Riemannian volume form is $dV = \sqrt{\det G_5}\,\sqrt{1 + \|\nabla r\|^2_{G_5^{-1}}}\,r^3\,dc \wedge d\mathrm{vol}_{S^3}$, and fibre integration (push-forward along $\pi : X_8 \to \mathcal{C}$) gives

$$\pi_* dV = 2\pi^2\,r(c)^3\sqrt{\det G_5}\,\sqrt{1 + \|\nabla r\|^2_{G_5^{-1}}}\,dc.$$

4. The chord $\delta^{(8)}$ of the 8S edition and the geodesic distance d_g of ds^2 satisfy $\delta^{(8)}(z_i, z_j) \leq d_g(z_i, z_j)$.

Proof. (1) Differentiate the embedding: $d\Phi = (dc,\ (\nabla r\cdot dc)\,u + r\,du)$. Then

$$\|d\Phi\|^2_{\mathcal{G}_8} = dc^\top G_5 dc + (\nabla r\cdot dc)^2\|u\|^2 + 2r(\nabla r\cdot dc)\,u\cdot du + r^2\|du\|^2.$$

On S^3, $\|u\|^2 = 1$ so $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$, killing the cross term, and $\|u\|^2 = 1$ in the second. Collecting, $(\nabla r\cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$, which gives the stated form.

(2) The matrix determinant lemma: for invertible A and vectors v, $\det(A + vv^\top) = \det A\,(1 + v^\top A^{-1}v)$. Apply with $A = G_5$, $v = \nabla r$.

(3) The volume form of a product-type metric is the product of the volume forms; the base factor contributes $\sqrt{\det(G_5 + \nabla r\nabla r^\top)}\,dc$, which is (2), and the fibre factor is the volume form of a round 3-sphere of radius r, namely $r^3 d\mathrm{vol}_{S^3}$. Pushforward integrates out the fibre, and $\mathrm{vol}(S^3) = 2\pi^2$.

(4) A geodesic from z_i to z_j is a curve in $X_8 \subset \mathbb{R}^9$; its ambient length is at least the ambient straight-line distance, which for a common fibre point is $\delta^{(8)}$ by 8S Theorem E(2), and the intrinsic length dominates the ambient length because the induced metric is the restriction of $\mathcal{G}_8$. ■

Theorem Θ-F — the projection contracts, and the coarea formula

Under (N):

1. $\Pi(\tau)$ is 1-Lipschitz, and for any C^1 function φ on $\mathbb{R}^3$, $\|\nabla(\varphi\circ\Pi)\| \leq \|\nabla\varphi\| \circ \Pi$.
2. $g^{(3)} \leq g^{(5)}$, so a separation visible after projection is real (8S Theorem F , unchanged by the limit).

3. **Coarea.** For Π a submersion and F integrable on $\mathbb{R}^9$,

$$\int_{\mathbb{R}^9} F\,\sqrt{\det(\Pi\Pi^\top)}\,dz = \int_{\mathbb{R}^3}\Big(\int_{\Pi^{-1}(x)} F\,d\mathcal{H}^6\Big)dx,$$

and under (N) $\sqrt{\det(\Pi\Pi^\top)}=1$, so the shadow of a distribution is its fibre integral.

Proof. (1) $\Pi\Pi^\top = I_3$ makes $\Pi^\top\Pi$ an orthogonal projector (8S Theorem F), so $\|\Pi v\|\le\|v\|$; the gradient statement is the chain rule $\nabla(\varphi\circ\Pi)=\Pi^\top(\nabla\varphi\circ\Pi)$ together with $\|\Pi^\top\|_2=1$. (2) is 8S Theorem F verbatim; it is an inequality between real numbers and does not involve the discretisation, so it passes to the limit unchanged. (3) is the coarea formula for Lipschitz submersions, with Jacobian factor $\sqrt{\det(\Pi\Pi^\top)}$, which (N) sets to 1. ■

Theorem Θ-G — the 8S Laplacian converges to an elliptic operator

Assume (R1) and (R2). Then for every c in the interior,

$$(\mathcal{L}_\sigma f)(c) \;=\; \mathcal{L}_\star f(c)+O(\sigma^2),\qquad \mathcal{L}_\star f=-\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla f)=-\tfrac12\operatorname{tr}(G_5^{-1}\nabla^2 f)-\frac{\nabla f^\top G_5^{-1}\nabla p}{p}.$$

Moreover $\mathcal{L}_\star$ is self-adjoint and positive semidefinite on $L^2(p^2dc)$, $\mathcal{L}_\star\mathbf{1}=0$, and on the compact $\mathcal{C}$ with Neumann conditions its spectrum is discrete, $0=\mu_1<\mu_2\le\mu_3\le\cdots\to\infty$, with

$$\mu_2=\min\Big\{\frac{\int\|\nabla\phi\|^2_{G_5^{-1}}\,p^2dc}{\int\phi^2p^2dc}\;:\;\phi\neq0,\ \int\phi\,p^2dc=0\Big\}.$$

Proof. Write $u=c'-c$ and $w(u)=\exp(-u^\top G_5u/2\sigma^2)$. Substitute $u=\sigma G_5^{-1/2}v$, so that $u^\top G_5u=\sigma^2\|v\|^2$ and $du=\sigma^5\det(G_5)^{-1/2}dv$. Write $Z=\int e^{-\|v\|^2/2}dv=(2\pi)^{5/2}$ and $\kappa=\sigma^5\det(G_5)^{-1/2}Z$. The Gaussian moments are

$$\int w\,du=\kappa,\qquad \int w\,u\,du=0,\qquad \int w\,uu^\top du=\sigma^2\kappa\,G_5^{-1},$$

the first-order moment vanishing by symmetry of w and the second following from $\int vv^\top e^{-\|v\|^2/2}dv=Z\,I$ conjugated by $G_5^{-1/2}$. Third moments vanish by symmetry; fourth moments are $O(\sigma^4\kappa)$.

Taylor-expand with (R1): $f(c+u)=f+\nabla f^\top u+\frac12u^\top\nabla^2f\,u+O(\|u\|^3)$ and $p(c+u)=p+\nabla p^\top u+O(\|u\|^2)$, all derivatives at c. Then the numerator of $I-\mathcal{A}_\sigma$ is

$$N=\int w(u)[f(c)-f(c+u)]p(c+u)\,du=-\Big[\underbrace{p\int w\,\nabla f^\top u\,du}_{=0}+\tfrac{p}{2}\int w\,u^\top\nabla^2f\,u\,du+\int w\,\nabla f^\top u\,\nabla p^\top u\,du\Big]+O(\sigma^4\kappa),$$

where the two surviving terms are the only ones of order $\sigma^2\kappa$ and the odd-order terms drop by symmetry. Evaluating with the moments,

$$N=-\sigma^2\kappa\Big[\tfrac{p}{2}\operatorname{tr}(G_5^{-1}\nabla^2f)+\nabla f^\top G_5^{-1}\nabla p\Big]+O(\sigma^4\kappa).$$

The denominator is $D=\int w\,p(c+u)\,du=p\,\kappa+O(\sigma^2\kappa)$. Dividing and multiplying by σ^{-2},

$$\mathcal{L}_\sigma f=\frac{N}{\sigma^2D}=-\tfrac12\operatorname{tr}(G_5^{-1}\nabla^2f)-\frac{\nabla f^\top G_5^{-1}\nabla p}{p}+O(\sigma^2).$$

That the displayed expression equals $-\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla f)$ is the product rule:

$$\nabla\cdot(p^2G_5^{-1}\nabla f)=p^2\operatorname{tr}(G_5^{-1}\nabla^2f)+2p\,\nabla p^\top G_5^{-1}\nabla f.$$

Self-adjointness, semidefiniteness, and the kernel are Theorem Θ-C(1)–(2). Discreteness of the spectrum on the compact $\mathcal{C}$ with Neumann conditions follows from the Rellich–Kondrachov compact embedding $H^1\hookrightarrow L^2$ and the spectral theorem for operators with compact resolvent. The variational formula for μ_2 is the Courant–Fischer min-max principle applied on the orthogonal complement of the constants. ■

Remark Θ-G (the density drift, and how to remove it). The limit is not the plain Laplace–Beltrami operator: it carries the drift $-p^{-1}\nabla f^{\top} G_5^{-1}\nabla p$ inherited from the sampling density. Normalising the kernel first, $\widetilde{W}_\sigma = W_\sigma/(D_\sigma(c)D_\sigma(c'))$, and only then forming degrees and the Laplacian, cancels p entirely and yields $-\frac{1}{2}\operatorname{tr}(G_5^{-1}\nabla^2\cdot)$. Entries whose input content is density-free invoke the normalised construction; entries whose content genuinely involves a sampling measure use the form as stated. Each per-entry proof says which.

Remark Θ-G (the boundary). (R2) is not cosmetic. Within $O(\sigma)$ of $\partial\mathcal{C}$ the kernel is truncated, the first moment no longer vanishes, and the expansion above acquires an $O(\sigma^{-1})$ term whose effect is to impose the Neumann condition $\partial_\nu f = 0$ in the limit. So $\mathcal{L}_\star$ on the penteract carries reflecting boundary conditions, which is why $\mu_1 = 0$ with constant eigenfunction rather than a Dirichlet $\mu_1 > 0$.

Remark Θ-G (the numerical check). With $d = 1$, $G_5 \to g = 2.3$, $f = \sin 2x + 0.3x^2$, $p = e^{-x^2/2} + 0.4$ evaluated at $x = 0.37$ by quadrature, the residual $|\mathcal{L}_\sigma f - \mathcal{L}_\star f|$ falls as 2.4×10^{-2}, 6.1×10^{-3}, 1.5×10^{-3}, 3.8×10^{-4} for $\sigma = 0.2, 0.1, 0.05, 0.025$ — a ratio of 4 per halving, confirming the $O(\sigma^2)$ rate.

Theorem Θ-H — the gate layer is a gradient system for a free energy

Let $m = 2\gamma - 1 \in (-1,1)^N$, let W be symmetric and H fully symmetric, and set

$$\mathcal{F}(m) = \sum_i \Big[\tfrac{1+m_i}{2}\log\tfrac{1+m_i}{2} + \tfrac{1-m_i}{2}\log\tfrac{1-m_i}{2}\Big] - \tfrac{1}{2}\sum_i b_i m_i - \tfrac{J_2}{4}\sum_{i,j} W_{ij}m_i m_j - \tfrac{J_3}{6}\sum_{i,j,k} H_{ijk}m_i m_j m_k.$$

1. $\nabla\mathcal{F}(m) = 0$ **iff** $m = \tanh(g)$ with $g = \frac{1}{2}b + \frac{J_2}{2}Wm + \frac{J_3}{2}\langle H, mm\rangle$, which is exactly the 8S gate fixed point $p = \sigma(h)$ under $m = 2p-1$, $g = h/2$.
2. $\mathcal{F}$ is strictly convex, hence the fixed point is unique, whenever $\nabla^2\mathcal{F} \succ 0$; a sufficient condition is $\frac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} < 1$, the 8S condition (C).
3. The gradient flow $\dot{m} = -\nabla\mathcal{F}(m)$ decreases $\mathcal{F}$ strictly off equilibria, so $\mathcal{F}$ is a Lyapunov function.

Proof. (1) Differentiate. The entropy term gives $\frac{1}{2}\log\frac{1+m_i}{1-m_i} = \operatorname{artanh}(m_i)$. The field term gives $-\frac{1}{2}b_i$. The pair term, using $W = W^{\top}$, gives $-\frac{J_2}{4}\cdot 2\sum_j W_{ij}m_j = -\frac{J_2}{2}(Wm)_i$. The triple term, using full symmetry of H, gives $-\frac{J_3}{6}\cdot 3\sum_{j,k} H_{ijk}m_j m_k = -\frac{J_3}{2}\sum_{j,k} H_{ijk}m_j m_k$. Setting the sum to zero gives $\operatorname{artanh}(m_i) = g_i$, i.e. $m = \tanh(g)$. Substituting $m = 2p-1$ and $g = h/2$ turns $\tanh(h/2) = 2\sigma(h) - 1$ into $p = \sigma(h)$, which is the 8S fixed point equation.

(2) The Hessian is $\nabla^2\mathcal{F} = \operatorname{diag}((1-m_i^2)^{-1}) - \frac{J_2}{2}W - J_3\langle H, m\rangle$. The first term is $\succeq I$ since $(1-m^2)^{-1} \geq 1$. By Gershgorin the remaining terms perturb the eigenvalues by at most $\frac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}}$ in modulus, so the stated bound forces $\nabla^2\mathcal{F} \succ 0$; strict convexity gives at most one critical point, and coercivity of the entropy at $m \to \pm 1$ gives at least one.

(3) $\frac{d}{dt}\mathcal{F}(m(t)) = \nabla\mathcal{F}\cdot\dot{m} = -\|\nabla\mathcal{F}\|^2 \leq 0$, with equality only at critical points. ■

Remark Θ-H . This upgrades 8S Theorem H from a contraction estimate to a *variational principle*: the gate is not merely a fixed point of some map, it is the minimiser of a free energy, and the 8S iteration is a descent scheme for it. When condition (C) fails, $\mathcal{F}$ loses convexity and acquires multiple minima — the bistability that entries of class D13 and D14 record.

Theorem Θ-I — the radius law is an exact differential form

Assume $B_5, s > 0$ and $r = r_{\max} B_5^{\alpha} s^{1/3}$.

1. $\omega := d\log r = \alpha\, d\log B_5 + \frac{1}{3}\, d\log s$ is an exact 1-form, hence closed, $d\omega = 0$, and its integral is path-independent.
2. In the factorised coordinates $(\log b(c_4), \dots, \log b(c_8), \log s)$, which are independent by construction,

$$\left.\frac{\partial\log r}{\partial\log B_5}\right|_s = \alpha, \qquad \left.\frac{\partial\log r}{\partial\log s}\right|_{B_5} = \tfrac{1}{3}, \qquad \left.\frac{\partial\log r}{\partial\log b(c_a)}\right|_{s,\, b(c_{a'})\, a'\neq a} = \alpha \quad (a = 4, \dots, 8),$$

so the elasticity with respect to each of the five bump factors is the same constant α. **The variables held fixed matter.** These are partial derivatives in the factorised coordinates, not directional derivatives in c: if s itself depends on c — as it does whenever the salience varies over the base — then the *total* derivative

along c_a is $\alpha\,\partial_a \log b + \frac{1}{3}\partial_a \log s$, and the second term does not vanish. Entries that report an elasticity must say which variables are held fixed, and the per-entry proofs do.

3. $\nabla r = r(\alpha \nabla \log B_5 + \frac{1}{3}\nabla \log s)$, which is the quantity entering the bundle metric of Θ-E.

Proof. (1) $\log r = \log r_{\max} + \alpha \log B_5 + \frac{1}{3}\log s$ is a well-defined C^1 function where the arguments are positive; ω is its differential, hence exact. Exact forms are closed ($d \circ d = 0$) and their line integrals depend only on endpoints (fundamental theorem of calculus for line integrals). (2) Read off the partial derivatives, using $\log B_5 = \sum_{a=4}^{8} \log b(c_a)$ so each summand carries coefficient α. (3) Multiply (1) by r and use $\nabla \log r = \nabla r / r$. ■

Theorem Θ-J — conservativity: the tower retracts

Let S_ℓ denote restriction to $\mathcal{C}_M$ together with the Riemann-sum quadrature $\int_{\mathcal{C}} F\,dc \mapsto \ell^5 \sum_n F(\ell n)$.

1. For $F \in C^2(\mathcal{C})$, $\left|\ell^5 \sum_n F(\ell n) - \int_{\mathcal{C}} F\,dc\right| = O(\ell^2)$ on the interior, so every integral operator of Definition 2.2 restricts to its 8S sum with $O(\ell^2)$ error.
2. Consequently $S_\ell \circ \mathsf{T}_{\mathrm{cal}} = \mathsf{T}_{8S} + O(\ell^2)$, and exactly T_{8S} when the quadrature is taken as the definition of the integral.
3. Composing with the 8S retraction, $\mathsf{T}^{(0)}_{8S} \circ S_\ell \circ \mathsf{T}_{\mathrm{cal}}[\mathsf{T}_{8S}[\mathcal{L}]] = \mathcal{L}$ for every input law $\mathcal{L}$ of the corpus.

Proof. (1) is the standard midpoint/Riemann error estimate: on each cell of side ℓ the error of the midpoint rule is $\frac{\ell^7}{24}\,\mathrm{tr}\,\nabla^2 F + O(\ell^8)$ per cell, and there are $O(\ell^{-5})$ cells, giving $O(\ell^2)$ in total. (2) Each substitution in Definition 2.2 replaces a lattice sum by the corresponding integral and a difference operator by the corresponding derivative; applying S_ℓ inverts each substitution, up to the quadrature error of (1) and, for $\mathcal{L}_\star$, up to the $O(\sigma^2)$ of Θ-G. (3) is (2) followed by 8S Theorem J, which states $\mathsf{T}^{(0)}_{8S} = \mathrm{id}$. ■

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4. The fourteen differential lemmas

The class Tk assigned in the 8S edition determines the calculus class Dk; the assignment is functorial, and the diagram $\mathsf{T}_{\mathrm{cal}} \circ \mathsf{T}_{8S}$ commutes with the retractions by Θ-J.

Lemma D1 — linear map → Fredholm integral operator and its generator

$$(\mathcal{K}_\sigma \psi)(c) = \int_{\mathcal{C}} \gamma(c) W_\sigma(c,c') \gamma(c')\, \mathsf{A}\, \psi(c')\, p(c')\, dc', \qquad \partial_\tau \psi = -\mathcal{L}_\star \psi.$$

Proof. The Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ of 8S Lemma T1 has the sum $\sum_j$ over nodes; Definition 2.2 replaces it by the integral against $p\,dc'$, giving a Fredholm operator of the second kind with continuous kernel, hence compact on L^2 by the Arzelà–Ascoli/Hilbert–Schmidt criterion $\iint |K|^2 < \infty$. Its generator in the bandwidth limit is $-\mathcal{L}_\star$ by Θ-G. Retraction: S_ℓ returns the finite sum (Θ-J(1)), and the 8S retraction returns $y = Ax$. ■

Lemma D2 — spectral statement → Sturm–Liouville problem and Rayleigh quotient

$$-\frac{1}{2p^2}\nabla \cdot (p^2 G_5^{-1} \nabla \phi_k) = \mu_k \phi_k, \qquad \mu_2 = \min_{\substack{\phi \neq 0 \\ \int \phi p^2 = 0}} \frac{\int \|\nabla \phi\|^2_{G_5^{-1}} p^2 dc}{\int \phi^2 p^2 dc}.$$

Proof. Θ-G supplies the operator, its self-adjointness, the discreteness of the spectrum, and the min-max characterisation. The dictionary of 8S Lemma T2 transfers term by term: stability $\leftrightarrow \mu_2 > 0$, relaxation time $\leftrightarrow 1/\mu_2$, threshold $\leftrightarrow \mu_2 = 0$, mode count $\leftrightarrow \dim\ker \mathcal{L}_\star$. Retraction: discretising the Rayleigh quotient on the lattice returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x / x^\top x$. ■

Lemma D3 — quadratic form → Dirichlet energy and the cone metric

$$\mathcal{E}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}} p^2 dc, \qquad ds^2 = dc^\top (G_5 + \nabla r \nabla r^\top) dc + r^2 d\Omega_3^2.$$

Proof. Θ-C(1) identifies $\mathcal{E}$ as the quadratic form of $\mathcal{L}_\star$ and Θ-C(3) gives the Euler–Lagrange equation as its first variation; Θ-E(1) gives the metric. Positivity, the surviving content of Sylvester's law, is Θ-C(2). Retraction: $\nabla \to$ difference quotients returns $x^\top \mathcal{L}_{8S} x$; $r \to$ constant returns $\delta^{(8)}$. ■

Lemma D4 — projection → Hilbert projection theorem and Gram volume

$$\langle e - P_{\mathcal{V}} e,\ v\rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta_{\mathrm{ind}} = \frac{\det \mathrm{Gr}(C_4, e)}{\det \mathrm{Gr}(C_4)(\|e\|^2 + \varepsilon)}, \qquad \nabla_e \|w\|^2 = 2w.$$

Proof. Θ-A(1)–(3) in full. The variational condition is the first-order condition for the least-squares functional, so the 8S normal-equation content becomes a stationarity statement. Retraction: on the lattice the inner products become finite sums and the Gram ratio becomes the 8S residual. ■

Lemma D5 — rank and null space → kernel dimension, index, and H^0

$$\dim \ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{connected components}\}, \qquad \mathrm{ind}\, \mathcal{L}_\star = \dim \ker \mathcal{L}_\star - \dim \mathrm{coker}\, \mathcal{L}_\star = 0.$$

Proof. Θ-C(2) gives $\ker \mathcal{L}_\star$ = locally constant functions, whose dimension is the number of connected components — the zeroth de Rham cohomology of $\mathcal{C}$. Self-adjointness gives coker ≅ ker, hence index zero, which is the analytic counterpart of the symmetry of the 8S Laplacian. Retraction: the lattice statement is 8S Lemma T5. ■

Lemma D6 — log-linear law → exact differential form and elasticity

$$d \log r = \alpha\, d \log B_5 + \tfrac{1}{3}\, d \log s, \qquad d(d \log r) = 0, \qquad \frac{\partial \log r}{\partial \log s} = \tfrac{1}{3}.$$

Proof. Θ-I(1)–(2). Exactness is what makes the law integrable and path-independent, which is the calculus content of the 8S statement that the logarithms add. Retraction: integrating the form between two lattice points returns the 8S difference of logarithms. ■

Lemma D7 — threshold → bifurcation and the implicit function theorem

$$\mathcal{G}(x, \lambda) = 0, \qquad \partial_x \mathcal{G} \text{ invertible} \Rightarrow x = x(\lambda) \text{ locally } C^1; \qquad \text{criticality} \iff \partial_x \mathcal{G} \text{ singular}, \quad \frac{d\mu_2}{d\lambda}\Big|_{\lambda_c} \neq 0.$$

Proof. The implicit function theorem gives a locally unique smooth branch exactly while the linearisation is invertible; a threshold is where invertibility fails, which for the spectral thresholds of 8S Lemma T7 is $\mu_2 = 0$ and for the gate threshold is loss of strict convexity of $\mathcal{F}$ (Θ-H(2)). Transversality $d\mu_2/d\lambda \neq 0$ makes the crossing simple, so the branch structure is that of a fold. Retraction: on the lattice these are the three 8S thresholds. ■

Lemma D8 — kernel → heat semigroup and its symbol

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0, \qquad W_\sigma \mathrel{\widehat{=}} e^{t\mathcal{A}_G}\big|_{t=\sigma_c^2/2}, \qquad \widehat{k_t}(\xi) = e^{-t\xi^\top G_5^{-1} \xi}.$$

Proof. Θ-D(1)–(3) in full: the Gaussian coupling kernel of the master law *is* a heat kernel, with the bandwidth as the square root of twice the diffusion time. Positive semidefiniteness, which the 8S edition obtained from Schoenberg's theorem, is here immediate from (3), since the Fourier transform of the kernel is positive. Retraction: sampling the kernel on the lattice returns W_{ij}. ■

Lemma D9 — metric → geodesic distance and the eikonal equation

$$d_g(z, z') = \inf_\Gamma \int_0^1 \|\dot\Gamma\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)} \leq d_g, \qquad g^{(3)} \leq g^{(5)}.$$

Proof. The first is the definition of Riemannian distance for the metric of Θ-E(1); the eikonal identity is the standard first-variation result that the distance function has unit gradient away from the cut locus; the chord bound is Θ-E(4); the projected ordering is Θ-F(2). Retraction: on the lattice, geodesics through common fibre points have length $\delta^{(8)}$. ■

Lemma D10 — Markov chain → Fokker–Planck equation and Wasserstein gradient flow

$$\partial_t \rho = \nabla\cdot\Big(G_5^{-1}(\tfrac{1}{2}\nabla\rho + \rho\nabla V)\Big), \qquad \rho_\infty \propto e^{-2V}, \qquad \partial_t\rho = \nabla\cdot\Big(\rho\,\nabla\frac{\delta\mathcal{F}}{\delta\rho}\Big).$$

Proof. The 8S chain $\widehat{P} = D^{-1}\Gamma W\Gamma$ is reversible with respect to π^{8S}; its continuum generator is the adjoint of $\mathcal{L}_\star$, and writing $\pi^{8S} \propto e^{-2V}$ turns the generator into the displayed drift-diffusion form. Stationarity: substituting $\rho = e^{-2V}$ makes the bracket $\frac{1}{2}\nabla e^{-2V} + e^{-2V}\nabla V = -e^{-2V}\nabla V + e^{-2V}\nabla V = 0$. The gradient-flow form is the Jordan–Kinderlehrer–Otto identification of Fokker–Planck as the steepest descent of the free energy $\mathcal{F}[\rho] = \int(\frac{1}{2}\rho\log\rho + \rho V)$ in the Wasserstein-2 metric. Retraction: discretising returns the reversible chain. ■

Lemma D11 — optimisation → Euler–Lagrange equation and KKT conditions

$$\delta\Big[\Delta_{8S} - \lambda(\|n\|_1 - M) - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}})\Big] = 0, \qquad \lambda,\nu \geq 0, \quad \lambda(\|n\|_1 - M) = \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}) = 0.$$

Proof. The 8S problem is a linear programme over a cross-polytope. Its continuum relaxation is a constrained variational problem; stationarity of the Lagrangian is the first-order necessary condition, and since the objective is concave and the feasible set convex, the KKT conditions are also sufficient. Complementary slackness records which constraint binds. Retraction: on the lattice the variation becomes a difference and the conditions become the LP optimality conditions. ■

Lemma D12 — convolution → Fourier transform and pseudodifferential symbol

$$\widehat{W_\sigma * \psi}(\xi) = \widehat{W_\sigma}(\xi)\,\widehat{\psi}(\xi), \qquad \widehat{W_\sigma}(\xi) = e^{-\sigma_c^2\xi^\top G_5^{-1}\xi/2}, \qquad \widehat{\partial_a\psi}(\xi) = i\xi_a\widehat{\psi}(\xi).$$

Proof. The convolution theorem for L^1 functions; the Gaussian transform of Θ-D(3) with $t = \sigma_c^2/2$; and the differentiation rule, obtained by integrating by parts in the transform integral. Together these say the coupling operator is a pseudodifferential operator whose symbol is a Gaussian, and its logarithm the elliptic symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$. Retraction: on the lattice torus the transform becomes the five-fold DFT $F_5 = F^{\otimes 5}$ of 8S Lemma T12. ■

Lemma D13 — nonlinear input → Fréchet derivative with a quantified remainder

$$\mathcal{N}(x_0+\xi) = \mathcal{N}(x_0) + D\mathcal{N}[x_0]\,\xi + R[\xi], \qquad \|R[\xi]\| \leq \tfrac{1}{2}\sup_{\|\zeta\|\leq\|\xi\|}\big\|D^2\mathcal{N}[x_0+\zeta]\big\|\,\|\xi\|^2,$$

with $D\mathcal{N}$ the two-block operator whose phase block is $-\mathcal{L}_\star$ and whose gate block is $\nabla^2\mathcal{F}$ of Θ-H(2). *Proof.* Taylor's theorem with integral remainder in a Banach space: $R[\xi] = \int_0^1 (1-t)D^2\mathcal{N}[x_0+t\xi][\xi,\xi]\,dt$, whence the bound. The phase block is Θ-G; the gate block is the Hessian of the free energy, which by Θ-H(1) has the fixed point as its critical point. Validity requires $\nabla^2\mathcal{F} \succ 0$, i.e. condition (C). Retraction: the blocks return the 8S Jacobian blocks. ■

Remark D13 . The calculus edition improves on the 8S edition here in one concrete respect: the remainder is *quantified*. The 8S image asserted a linearisation; the calculus image bounds its error by a second-derivative supremum, so an entry's *Linearised* status now comes with a modulus of validity rather than a warning.

Lemma D14 — obstruction → vanishing first variation

$$\delta\Delta_{8S}[e] = 0 \ \text{ for every admissible variation} \qquad \Longleftrightarrow \qquad (I - P_{\mathcal{V}})e = 0 \qquad \Longrightarrow \qquad \nabla_e\eta_{\mathrm{ind}} \equiv 0.$$

Proof. By Θ-A(3), $\nabla_e\|w\|^2 = 2w = 2(I-P_{\mathcal{V}})e$, which vanishes identically exactly when $e \in \mathcal{V}$. In that case the score Δ_{8S} has zero Gâteaux derivative in every admissible direction, so the entry carries no first-order information: there is nothing to differentiate, and the calculus image is the statement that the variation vanishes. The determinant form is $\det \mathrm{Gr}(C_4, e) = 0$ by Θ-A(4): the parallelepiped is degenerate and has zero 5-volume. Retraction: this is the 8S rejection certificate. ■

Remark D14 . The obstruction is preserved a third time. In the LA edition it was "no faithful linear form"; in the 8S edition, "$\eta_{\mathrm{ind}} < \varepsilon_{\mathrm{ind}}$, no new dimension"; here, "the first variation vanishes identically, the Gram volume is zero". Three vocabularies, one fact.

5. The per-entry proof schema

Every entry in `01`–`80` carries a proof with exactly six steps.

- **(Q1) Class and differential signature.** The 8S image is exhibited, its class Tk recalled, the calculus class Dk assigned, and the order ord $\in \{0, 1, 2\}$ of the *original* equation recorded — since an equation already carrying derivatives needs less smoothing to have a calculus image.
- **(Q2) Image.** The calculus output equation is displayed.
- **(Q3) Regularity.** The hypotheses of the relevant lemma are verified for this entry: which of (R1), (R2), (R3), condition (C), and the normalisation (N) are needed, and what fails without them.
- **(Q4) Derivation.** The image is derived by the named calculus operation — Taylor expansion with Gaussian moments, integration by parts, the chain rule, separation of variables, the first variation, or the convolution theorem.
- **(Q5) Retraction through the tower.** Θ-J is applied: S_ℓ returns the 8S image, and the 8S retraction returns the Linear Algebra equation. Both steps are checked.
- **(Q6) Admission.** The admission rule is restated as a first-order condition.

A proof consisting of a regularity check, an application of a fully proved lemma, an explicit derivation, a two-step limit verification, and a scope statement is a complete proof. The lemmas it invokes are proved in §3 and §4 without appeal to anything outside standard real analysis, measure theory, and the classical results cited by name (Hilbert projection theorem; Schur determinant formula; matrix determinant lemma; Picard–Lindelöf; the divergence theorem; Courant–Fischer min-max; Rellich–Kondrachov; the coarea formula; Taylor's theorem with integral remainder; the implicit function theorem; the convolution theorem; Jordan–Kinderlehrer–Otto).

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6. Scope and honest limits

The bandwidth limit is a limit. Θ-G holds as $\sigma \to 0$ with an $O(\sigma^2)$ rate under (R1). At fixed σ the 8S operator is *not* an elliptic operator; it is a bounded integral operator whose spectrum accumulates at 0 rather than diverging. Entries whose content depends on the high end of the spectrum inherit this caveat, and it is stated in their (Q3).

The density drift is real. The limit is a weighted Laplacian, not the Laplace–Beltrami operator, unless the kernel is renormalised first (Remark Θ-G). Which construction an entry uses is stated in its (Q4).

The boundary imposes Neumann conditions. The penteract is compact with boundary, and the limit operator carries reflecting conditions (Remark Θ-G). Statements about Dirichlet spectra do not transfer without modification.

Convexity can fail. Θ-H(2) gives uniqueness only under condition (C). Beyond it the free energy has several minima, and the entries recording bistability or hysteresis sit there deliberately.

Obstructions are still obstructions. D14 entries have vanishing first variation. Differentiating them produces zero, not insight; the calculus edition reports that, and does not manufacture a differential equation where the corpus records that none exists.